

ESABCC MethodHub — Indicator Database

Post-report data update & source verification sheet

Generated 2026-06-02. This sheet documents every **existing** indicator (the 51 “ESABCC report” indicators rebuilt from the 2024 report *Towards EU climate neutrality*) and the result of a web search for **newer data published after the report**. Newly added values are flagged **afterReport** in the data file and render in orange (dashed extension, “NEW” badge) in the Indicator Database chart/table. **Please fact-check each new value against the linked source.**

Summary: 30 new data points added across 17 of 51 indicators. The remaining indicators were left unchanged — the reason (scope mismatch, source not machine-retrievable in this environment, or vintage conflict) is given in the Notes column.

Method & caveats. Direct fetching of Eurostat/EEA data tables (JSON API and data-viewer CSV) was blocked in the build environment (HTTP 403 / host-allowlist), so values were sourced from official Eurostat/EEA news releases and Statistics-Explained pages surfaced via web search, then unit-converted (Mtoe×11.63→TWh; PJ÷3.6→TWh) and cross-checked against the stored last value. Shares are stored as fractions. **Confidence:** high = figure stated verbatim in the cited source on a matching scope; medium = one year derived (e.g. via a stated % change) or built from rounded components; low = not added. Two finance series (F1, F3) have a **vintage caveat**: the publisher has since revised the report’s 2022 base, so the new 2023 point sits on a slightly revised baseline.

Code	Indicator (unit)	Report→	New points (year = value)	Source (click to open)	Conf.	Notes
O1	Total EU GHG emissions (European Climate Law scope) (Mt CO ₂ eq)	2022 3333.9	— none added	EEA – Total net GHG emission trends and projections https://www.eea.europa.eu/en/analysis/indicators/total-greenhouse-gas-emission-trends	low	PENDING — see own-fetch attempt. EU domestic excl-LULUCF fell 7.4% in 2023; net incl-LULUCF 2908 Mt; O1 gross+intl-transport exact 2023 not confirmed by agent.
O2 (PEC)	Primary energy consumption (EU-27) (TWh)	2021 15248.0	2022 = 14653.8, 2023 = 14083.9, 2024 = 13979.3	Eurostat – EU primary energy consumption decreased in 2024 (PEC2020-2030) https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-2025-1215-2	high	PEC2020-2030 metric, Mtoe×11.63: 2022=1260,2023=1211,2024=1202. Anchor 2021=1311 Mtoe→15247=stored 15248.
O2 (FEC)	Final energy consumption (EU-27) (TWh)	2021 11263.0	2022 = 10711.2, 2023 = 10397.2, 2024 = 10467.0	Eurostat – EU energy consumption 2024 (FEC2020-2030) https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-2025-1215-2	high	FEC2020-2030 metric, Mtoe×11.63: 2022=921,2023=894,2024=900. Anchor 2021=968→11258=stored 11263.
O3	Gross inland energy consumption (EU-27) (TWh)	2021 16534.0	2022 = 16241.1, 2023 = 15575.3	Eurostat – Energy statistics overview (gross available energy) https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Energy_statistics_-_an_overview	medium	Gross available energy 2023=56071 PJ (–4.1% vs 2022); 2022 derived 58468 PJ. PJ÷3.6. 2022 derived (medium).
E1	Energy supply GHG emissions (Mt CO ₂ eq)	2022 924.0	— none added	EEA – EU energy sector leads emission cuts in 2023 https://www.eea.europa.eu/en/newroom/news/greenhouse-gas-emissions-in-2023	low	PENDING. Energy supply –19% 2022→2023 (would give ~748 from 924) but no exact absolute.
E2 (fossil)	Fossil share of EU electricity mix (%)	2022 0.3876	— none added	Ember European Electricity Review 2025 https://ember-energy.org/latest-insights/european-electricity-review-2025/	low	Could not retrieve 4-decimal Eurostat nrg_bal_pch value. Search indicates ~32% (2023), ~29% (2024) but not at stored precision. Left out.
E2 (RES)	Non-biomass renewable share of EU electricity mix (%)	2022 0.3376	— none added	Ember European Electricity Review 2025 https://ember-energy.org/latest-insights/european-electricity-review-2025/	low	Non-biomass RES share needs component breakdown at 4-decimal precision; not reliably sourced. Left out.
E3	Average GHG intensity of EU electricity (g CO ₂ eq/kWh)	2022 251.0	— none added	EEA GHG emission intensity of electricity generation https://www.eea.europa.eu/en/analysis/indicators/greenhouse-gas-emission-intensity-of-1	low	Possible series/scope mismatch (search ~242 g 2023 vs stored 251 in 2022 on different basis). Left out to avoid mixing series.
E4a	Annual solar PV capacity additions (EU) (GW/yr)	2021 25.7	2023 = 56, 2024 = 65.5	SolarPower Europe EU Market Outlook for Solar Power https://www.solarpowereurope.org/insights/outlooks/eu-market-outlook-for-solar-power-2024-2028	medium	EU-27 new grid-connected solar PV: 2023=56 GW, 2024=65.5 GW. Matches stored series (2021=25.7). Read via search summaries (WebFetch 403).

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E4b/c	Annual wind capacity additions (EU) (GW/yr)	2021 10.72	2023 = 16.2, 2024 = 12.9	WindEurope - Wind energy in Europe 2024 statistics https://windeurope.org/intelligence-platform/product/wind-energy-in-europe-2024-statistics-and-the-outlook-for-2025-2030/	medium	EU-27 new wind installs: 2023=16.2 GW (13.3 onshore+2.9 offshore), 2024=12.9 GW. Matches stored series (2021=10.72).
E5	Electrification rate of final energy use (%)	2021 0.2275	— none added	Eurostat electricity overview https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Electricity_production_consumption_and_market_overview	low	No exact electricity-share fraction for 2022/2023 at stored precision. Left out.
E6	Energy-related methane emissions (EU-27) (Mt CO ₂ eq)	2021 69.91	— none added	EEA https://www.eea.europa.eu/en/analysis/indicators/total-greenhouse-gas-emission-trends	low	PENDING. Energy CH4 only to 2020 in accessible material.
I1	Industrial GHG emissions (Mt CO ₂ eq)	2022 691.3	— none added	EEA https://www.eea.europa.eu/en/analysis/indicators/total-greenhouse-gas-emission-trends	low	PENDING. Manufacturing ~7% 2023.
I3	Circular material use rate (EU) (%)	2021 0.117	2022 = 0.115, 2023 = 0.118	Almost 12% of materials in the EU came from recycling - Eurostat (13 Nov 2024) https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-2024-1113-1	high	Eurostat cei_srm030, EU-27. 2022=11.5%, 2023=11.8%. Stored as fraction to match 2021=0.117.
I4 (steel)	GHG intensity of EU steel production (t CO ₂ /t)	2021 0.9623	— none added	—	low	No citable source on the stored custom methodology (EEA emissions / EU-27 production). Omitted to avoid methodological mismatch.
I4 (cement)	GHG intensity of EU cement production (t CO ₂ /t)	2021 0.6038	— none added	—	low	No citable source matching stored basis; available figures are clinker-level or global. Omitted.
I4 (chemicals)	GHG intensity of EU base organic chemicals (t CO ₂ /t)	2022 0.767	— none added	—	low	No credible EU-27 base-organic-chemicals intensity for 2023 on stored basis. Omitted.
I5	Industrial final energy consumption (EU) (TWh)	2021 2795.2	2022 = 2627.8, 2023 = 2497.2	Eurostat – Final energy consumption in industry https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Final_energy_consumption_in_industry_-_detailed_statistics	medium	Industry FEC 2023=8990 PJ (–5.3% vs 2022); 2022=9460 PJ. PJ÷3.6. Anchor 2021=2795.2.
I6	Electricity share of industrial final energy use (%)	2021 0.3322	2022 = 0.333, 2023 = 0.326	Eurostat – Final energy consumption in industry detailed statistics https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-2024-0513-1	high	Electricity share of industry FEC: 2022=33.3%→0.333, 2023=32.6%→0.326. Anchor 2021=0.3322.
T1	Transport GHG emissions (EU) (Mt CO ₂ eq)	2022 906.6	— none added	EEA – GHG emissions from transport https://www.eea.europa.eu/en/analysis/indicators/greenhouse-gas-emissions-from-transport	low	PENDING. Domestic transport ~1% 2023 → ~898-900 from 906.6.
T2a	Total passenger transport demand (Gpkm)	2020 4435.8	— none added	EU Transport in Figures — Statistical Pocketbook 2024 https://transport.ec.europa.eu/facts-funding/studies-data/eu-transport-figures-statistical-pocketbook/statistical-pocketbook-2024_en	low	DG MOVE pocketbook table not fetchable (403); only-accessible EEA series uses broader scope incompatible with 4435.8 baseline. Left out to avoid scale mismatch.
T2b	Total freight transport demand (Gtkm)	2020 2254.0	— none added	EU Transport in Figures — Statistical Pocketbook 2024 https://transport.ec.europa.eu/facts-funding/studies-data/eu-transport-figures-statistical-pocketbook/statistical-pocketbook-2024_en	low	Pocketbook freight table 403-blocked; EEA freight figure includes extra-EU maritime (3471 Gtkm 2022) ≠ 2254.0 baseline. Left out.

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T3a	Road share of motorised inland passenger transport (%)	2020 0.865	— none added	EU Transport in Figures — Statistical Pocketbook 2024 https://transport.ec.europa.eu/facts-funding/studies-data/eu-transport-figures-statistical-pocketbook/statistical-pocketbook-2024_en	low	Modal-split table 403-blocked; EEA car-share-of-all-pkm definition differs from road-share-of-inland-motorised (0.865). Left out.
T3b	Intra-EU air passenger transport (Gpkm)	2020 177.9	— none added	EU Transport in Figures — Statistical Pocketbook 2024 https://transport.ec.europa.eu/facts-funding/studies-data/eu-transport-figures-statistical-pocketbook/statistical-pocketbook-2024_en	low	Intra-EU air table 403-blocked; EEA total-aviation figure mixes intra/extra-EU. Left out.
T4	Average CO ₂ intensity of new passenger cars (WLTP) (g CO ₂ /km)	2022 108.2	2023 = 106.4	[EU SDG 13_31] Average CO ₂ emissions per km from new passenger cars — EEA https://www.eea.europa.eu/en/data-hub/datahubitem-view/5d252092-d328-40d8-bca2-c0734bd6143b	high	EU-27 WLTP 2023 = 106.4 gCO ₂ /km (–1.6% vs 108.2 in 2022). Matches stored EU-27 series. From EEA search snippet (page 403 to direct fetch) — confirm exact decimal.
T5a	ZEV share of new passenger car registrations (%)	2022 0.1341	2023 = 0.146, 2024 = 0.136	ACEA — New car registrations 2024; battery-electric 13.6% market share https://www.acea.auto/pc-registrations/new-car-registrations-0-8-in-2024-battery-electric-13-6-market-share/	high	BEV-only share of EU new car registrations: 2023=14.6% (0.146), 2024=13.6% (0.136). Consistent with BEV-only baseline 2022=0.1341. 2023 src: acea.auto/.../14-6-market-share .
T5b	Zero-emission lorries in the EU fleet (vehicles)	2022 3794.0	— none added	European Alternative Fuels Observatory — EU27 road https://alternative-fuels-observatory.ec.europa.eu/transport-mode/road/european-union-eu27	low	No reliable zero-emission heavy lorry figure retrievable (EAFO/ACEA 403). Left out.
T6a	Fossil share of transport energy use (%)	2021 0.9344	— none added	Eurostat – Energy consumption for transport 2023 https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-2025-1128-1	low	Only road-transport fuel mix retrievable (not all-modes). Left out.
T6b	Use of first-generation (food-crop) biofuels in transport (TWh)	2015 118.1	— none added	SHARES / Eurostat https://www.eea.europa.eu/en/data-hub/datahubitem-view/3b7fe76c-524a-439a-bfd2-a6e4046302a2	low	Not researched in this pass (series last updated 2015; SHARES food-crop biofuel split requires manual extraction). Recommend manual pull.
B1	Residential + tertiary building GHG emissions (Mt CO ₂ eq)	2022 481.8	— none added	EEA https://www.eea.europa.eu/en/analysis/indicators/total-greenhouse-gas-emission-trends	low	PENDING. Residential+services –8% 2023 → ~443 from 481.8.
B2	Final energy consumption in buildings (TWh)	2021 4878.7	2022 = 4213.6, 2023 = 4037.9	Eurostat – Energy use in EU households + services https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-2025-0625-2	medium	Residential+services FEC: 2022=15169 PJ, 2023=14537 PJ (PJ÷3.6). Households rounded to 3 s.f., hence medium.
B5a	Fossil share of residential energy mix (%)	2021 0.4558	2023 = 0.42	Eurostat – Energy use in EU households 2023 https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-2025-0625-2	medium	Residential 2023 fossil = gas 29.5%+oil 10.3%+solid 2.2% = 42.0%→0.420. Anchor 2021=0.4558.
B5b	Fossil share of tertiary energy mix (%)	2021 0.3874	2023 = 0.322	Eurostat – Energy use in services sector 2023 https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-2025-0717-1	medium	Services 2023 fossil = gas 26.0%+oil 6.2% = 32.2%→0.322. Anchor 2021=0.3874.
B6	EU stock of installed heat pumps (million units)	2022 19.79	— none added	EHPA European heat pump market and statistics report 2024 https://ehpa.org/news-and-resources/press-releases/heat-pump-sales-14-times-greater-in-lead-countries/	low	EHPA 2023 stock ~24 million but across 21 European countries (broader than EU-27 stored 2022=19.79). Scope mismatch — NOT added; candidate 2023~24.0 if EU-27 table confirmed.

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A1	Agricultural non-CO ₂ emissions (Mt CO ₂ eq)	2022 360.9	— none added	EEA – GHG emissions from agriculture https://www.eea.europa.eu/en/analysis/indicators/greenhouse-gas-emissions-from-agriculture	low	PENDING. Agri non-CO ₂ –1.2% 2023 → ~356.6 from 360.9.
A3 (use)	Total fertiliser nitrogen use (EU) (Mt N)	2021 15.01	— none added	Eurostat - Use of fertilisers in EU agriculture down 10% in 2022 https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-2024-0628-1	low	2022 mineral N=8.9 Mt, +manure ~17.3 Mt — neither reconciles with stored 2021=15.01. Definitional mismatch; left out.
A3 (NUE)	Nitrogen use efficiency in EU agriculture (%)	2020 49.03	— none added	—	low	No credible 2021/2022 EU-27 NUE value found. Left out.
A7	Agricultural products used as bioenergy feedstock (Mt fresh)	2021 23.89	— none added	JRC - Biomass production, supply, uses and flows in the EU (2023) https://publications.jrc.ec.europa.eu/repository/bitstream/JRC132358/JRC132358_01.pdf	low	JRC reports multi-year averages, no single 2022 value matching 23.89. Left out.
L1	LULUCF net GHG balance (Mt CO ₂ eq)	2022 -244.0	2023 = -198.0	EEA – GHG emissions from LULUCF in Europe https://www.eea.europa.eu/en/analysis/indicators/greenhouse-gas-emissions-from-land	medium	EU-27 LULUCF net sink ~-198 Mt CO ₂ e in 2023. Vintage caveat: stored 2022=-244 from earlier vintage.
L3	Annual afforested area (thousand ha/yr)	2021 326.8	— none added	EEA - GHG emissions from LULUCF in Europe https://www.eea.europa.eu/en/analysis/indicators/greenhouse-gas-emissions-from-land	low	CRF activity-data sub-line not retrievable; no exact 2022 value vs 326.8. Left out.
L4	Annual deforested area (thousand ha/yr)	2021 -114.5	— none added	EEA - GHG emissions from LULUCF in Europe https://www.eea.europa.eu/en/analysis/indicators/greenhouse-gas-emissions-from-land	low	CRF sub-line not retrievable; no exact 2022 value vs -114.5. Left out.
L5	Annual settlement area increase (net land take proxy) (thousand ha/yr)	2021 142.2	— none added	EEA - GHG emissions from LULUCF in Europe https://www.eea.europa.eu/en/analysis/indicators/greenhouse-gas-emissions-from-land	low	Net land take sub-line not retrievable; no exact 2022 value vs 142.2. Left out.
L6	Forest land living-biomass carbon sink (Mt CO ₂ eq)	2021 231.8	— none added	EEA - Europe's land carbon sink declines https://www.eea.europa.eu/en/newroom/news/europes-land-carbon-sink-declines-but-its-potential-stays-high	low	Forest living-biomass sub-pool not isolable from aggregates; left out.
L7	Non-forest LULUCF net GHG emissions (Mt CO ₂ eq)	2021 97.44	— none added	EEA - GHG emissions from LULUCF in Europe https://www.eea.europa.eu/en/analysis/indicators/greenhouse-gas-emissions-from-land	low	Non-forest LULUCF ~94 Mt (2022) per narrative paraphrase, too imprecise vs 97.44 baseline. Left out.
L8	Total bioenergy use (EU-27) (TWh)	2021 1795.7	— none added	EurObserv'ER / IEA Bioenergy https://www.eurobserv-er.org/	low	Only a bioenergy component retrievable; total requires approximate share division. Left out.
F1	EU fossil-fuel subsidies (FFST) (billion EUR)	2022 122.0	2023 = 111.0	Energy subsidies report shows progress in 2023 — European Commission COM(2025)17 https://energy.ec.europa.eu/news/energy-subsidies-report-shows-progress-2023-2025-01-29_en	high	EU fossil-fuel subsidies fell to €111 bn in 2023. VINTAGE CAVEAT: this report revised 2022 to €136 bn (stored 2022=122.0 is an earlier vintage), so 122→111 understates the true fall.
F2	EU green bond issuance (billion EUR)	2022 243.0	— none added	—	low	No clean EU-27 total green-bond issuance in €bn matching stored 243.0 methodology. Left out.
F2 (share)	Green share of EU bond issuance (%)	2022 0.0674	2023 = 0.053, 2024 = 0.069	Green bonds in Europe — EEA (8th EAP indicator) https://www.eea.europa.eu/en/analysis/indicators/green-bonds-8th-eap	medium	EEA green-bond share: 2023=5.3% (0.053), 2024=6.9% (0.069). Stored 2022=0.0674; EEA vintage may differ slightly.

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F3	EU R&D expenditure (GERD) (% of GDP)	2022 2.27	2023 = 2.22	EU spent €381.4 billion on R&D in 2023 — Eurostat (rd_e_gerdtot) https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-2024-1211-2	high	EU-27 GERD/GDP = 2.22% in 2023. VINTAGE CAVEAT: current Eurostat reports 2022=2.21% (stored 2.27 is an earlier vintage).
F4	Climate-related share of EU patent filings (%)	2019 11.94	— none added	—	low	No EU-27 climate-patent share on same methodology newer than 2019 (11.94%). Left out.
F5	Cleantech investment in the EU (% of GDP)	2022 0.0006	— none added	Cleantech for Europe — Cleantech Annual Briefing 2023 https://www.cleantechforeurope.com/publications/cleantech-annual-briefing-2023	low	€11.1 bn EU cleantech VC in 2023; %-of-GDP not published by source. Derived 0.00065=0.0006 (same as 2022). Too derived — NOT added; candidate only.

Rows shaded orange received new post-report data. Unshaded rows were intentionally left unchanged for the reason in Notes — most often because the exact figure on the indicator's precise scope lives only in the EEA GHG data viewer / Eurostat JSON API, which could not be machine-read here; those should be pulled manually from the linked source.